

15.455x Mathematical Methods of Quantitative Finance

Week 1: Probability

Paul F. Mende
MIT Sloan School of Management

Finance at MIT

Where ingenuity drives results

Random variables, distributions, and moments

Random variables

- A **random variable** is a function which assigns a number to **events** in the **sample space**. (A better name might be "random-valued function on the sample space.")

$$X = \begin{cases} 1, & \text{heads} \\ -1, & \text{tails} \end{cases} \quad Z = \begin{cases} 1, & \text{success} \\ 0, & \text{failure} \end{cases}$$

$$Y = \{\text{sum of two dice}\}$$

- We describe the **probability of an outcome** in terms of the probability of a random variable taking a given value:

$$P(X = 1) = 1/2, \quad P(X^2 = 1) = 1,$$

$$P(Y = 2) = 1/36, \quad P(Y = 3) = 2/36, \dots$$

$$P(Z = 1) = p, \quad P(Z = 0) = q, \quad (\text{where } p + q = 1)$$

Continuous random variables

- Consider choosing a random number between 0 and 1, where all values are equally likely.
- Since there are an (uncountably) **infinite** number of values, the probability of any given value is zero:

$$\text{Prob}(X = x_0) = 0$$

- Does **not** mean event is impossible. Points on a line have "**measure zero.**"
- Instead, ask for probability to lie within a given range, e.g.,

$$\text{Prob}(a < X < b) = \int_a^b p(x)dx$$

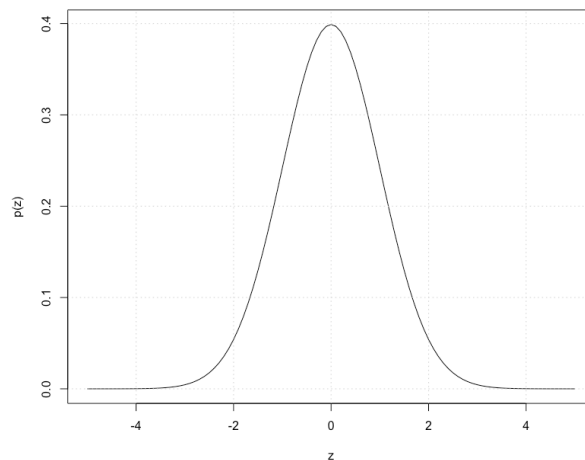
Probability distributions

- More generally, a **probability distribution** satisfies

$$p(x) \geq 0,$$

$$\sum_k p(x_k) = 1, \quad x_k \text{ discrete}$$

$$\int_{-\infty}^{\infty} p(x) dx = 1, \quad x \text{ continuous}$$



Cumulative distribution function

- The **cumulative distribution function**, or CDF, gives the probability that X is less than or equal to a given value.

$$F(x) = \text{Prob}(X \leq x) = \int_{-\infty}^x p(x') dx'$$

- It contains (nearly) the same information as the probability density, since

$$p(x) = \frac{dF(x)}{dx}$$

- The CDF is often easier to approximate from empirical data and it is useful since

$$\text{Prob}(a < X < b) = F(b) - F(a)$$

Change of variable

- Suppose we wish to change variables, shift the distribution, or consider functions of the random variable. If $x \rightarrow y = y(x)$ then the density in terms of the new variable is given by

$$p(x)dx = g(y)dy$$

which preserves the normalization condition. (This assumes y is an increasing function of x . If not, an absolute value is needed for positivity; and further care is needed if y has critical points.)

$$p(x)dx = g(y)dy$$

$$g(y) = \frac{p(x)}{|dy/dx|}$$

Expectations and moments

- The probability distribution defines weighted averages over the sample space, where the weight of each event is equal to its probability. These are called **expected values**.
- For the discrete case,

$$E[f(X)] = \sum_{k=1}^n f(x_k)p(x_k)$$

while for the continuous case,

$$E[f] = \int_{-\infty}^{\infty} f(x)p(x)dx$$

Mean of a distribution

- The **mean** of the distribution is simply the expectation of the random variable itself:

$$\mu \equiv E[X] = \overline{X} = \langle X \rangle = \begin{cases} \sum_k x_k p(x_k) \\ \int x p(x) dx \end{cases}$$

- In the case of an infinite sample space, whether continuous or discrete, the mean is not guaranteed to exist since the integral or the sum might not converge.

Moments of a distribution

- The **moments** of a distribution are the expectation of powers of the random variable itself.

$$\mu_\ell \equiv E[X^\ell] \equiv \langle X^\ell \rangle = \begin{cases} \sum_k x_k^\ell p(x_k) \\ \int x^\ell p(x) dx \end{cases}$$

- If all the moments are known – and if they exist – they can be used to get the expectation of other functions using the **linearity** of the expectation operator

$$\begin{aligned} E[cf(X)] &= c E[f(X)], \\ E[f(X) + g(X)] &= E[f(X)] + E[g(X)] \end{aligned}$$

Variance and standard deviation

- Of particular interest is the second moment, in combination with the mean, defining the **variance**:

$$\begin{aligned}
 \sigma^2 = \text{Var}(X) &\equiv \text{E} [(X - \mu)^2] \\
 &= \text{E} [X^2 - 2\mu X + \mu^2] \\
 &= \text{E} [X^2] - 2\mu \text{E} [X] + \mu^2 \\
 &= \text{E} [X^2] - \mu^2 \\
 &= \text{E} [X^2] - \text{E} [X]^2
 \end{aligned}$$

- The standard deviation, which is the square root of the variance, has the **same units** as the random variable (e.g., rate of return, dollars, etc.)

Higher moments characterize properties of a distribution

Variance – dispersion measure based on second moment

$$\sigma^2 \equiv E[(X - \mu)^2] = \int (x - \mu)^2 p(x) dx$$

Skewness – asymmetry parameter based on 3rd moments; **dimensionless**
– normalized cumulant

$$s \equiv \frac{E[(X - \mu)^3]}{\sigma^3} = E\left[\left(\frac{X - \mu}{\sigma}\right)^3\right]$$

Kurtosis – measure of tail "weights" in terms of 4th moments; zero for Gaussian, bounded below by -1.

$$\kappa \equiv \frac{E[(X - \mu)^4]}{\sigma^4} - 3$$

Covariance and correlation

- For any **two** random variables, not necessarily independent **or** identically distributed, their covariance is defined as

$$\text{Cov}(X, Y) \equiv \text{E}[(X - \mu_x)(Y - \mu_y)] = \text{E}[XY] - \mu_x \mu_y$$

- The correlation is proportional to the covariance,

$$\rho(X, Y) = \text{Corr}(X, Y) \equiv \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \text{E}\left[\left(\frac{X - \mu_x}{\sigma_x}\right)\left(\frac{Y - \mu_y}{\sigma_y}\right)\right]$$

- Dividing the covariance by the standard deviations makes the correlation a pure number, and

$$-1 \leq \rho(X, Y) \leq +1$$

Covariance

- If the variables are independent, then the covariance will vanish. However the converse is not true.
- Example: Let Y be a function of X , so it is completely dependent:

$X = \pm a$ each with probability p_1 , $\pm b$ with probability p_2

$Y = X^2 = a^2$ with probability $2p_1$, b^2 with probability $2p_2$

$\mu_x = 0, \quad \mu_y = E[Y] = 2p_1a^2 + 2p_2b^2$

$$\begin{aligned} \text{Cov}(X, Y) &\equiv E[(X - \mu_x)(Y - \mu_y)] = E[XY] - \mu_x\mu_y = E[XY] \\ &= p_1 \cdot a \cdot a^2 + p_1(-a) \cdot a^2 + p_2 \cdot b \cdot b^2 + p_2(-b) \cdot b^2 \\ &= 0 \end{aligned}$$

- Vanishing covariance does not imply independence.

Random variables

Summary

- "**Random variables**" are functions that assign a **number** to events in the sample space. They can be discrete, continuous, or a mix of both.
- The probability distribution is positive and sums to one.
- Expected values, or expectations, are weighted averages that use the probabilities as the weights.
- The **moments** of a distribution, such as the mean, are expectations of various powers of a random variable. They are numbers, not functions.
- Variance, skewness, and kurtosis are simple functions of the moments that characterize the shape of the probability distribution.
- When there are multiple random variables, their **covariance** and **correlation** are also computed as expectations.

Common distributions

Uniform distribution

- Standard form: $p(x) = \begin{cases} 1, & x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$,

$$\text{Prob}(a < X < b) = \int_a^b p(x) dx = (b - a)$$

$$F(x) = \int_{-\infty}^x p(x') dx' = x$$

- Moments:

$$\mu = \int_{-\infty}^{\infty} x p(x) dx = \int_0^1 x dx = \frac{1}{2}$$

$$\mu_\ell = \int_0^1 x^\ell dx = \frac{1}{\ell + 1}$$

$$\sigma^2 = \int_0^1 \left(x - \frac{1}{2}\right)^2 dx = \frac{1}{12}$$

Binomial distribution

- The **binomial distribution** provides a model for any observation or experiment that has two possible outcomes, "success" or "failure."
- Examples:
 - Coin toss: Heads vs. Tails
 - Roulette: 19 vs. {all other numbers}
 - Bond default prior to time T vs. no default
 - One-day trading loss $> \$100M$ vs. $(P/L > -\$100M)$

Binomial distribution

- Identify events by total number of successes vs. failures, independent of the order in which they occur.
- **Two parameters:** probability of success p , number of trials n .

$$f(k; n, p) = \binom{n}{k} p^k q^{n-k}, \text{ where}$$

$q = 1 - p$ is probability of “failure,”

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \binom{n}{n-k}$$

Binomial distribution: moments

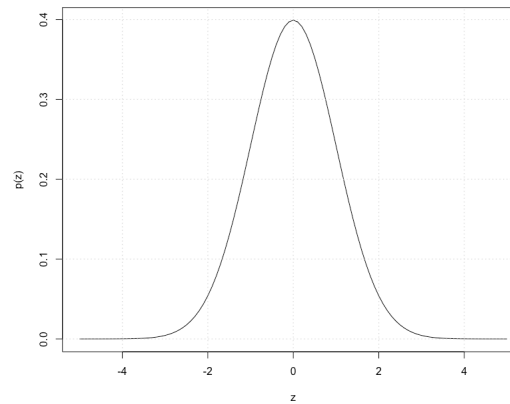
- Mean value (the hard way):

$$\begin{aligned}
 \mu = E[X] &= \sum_{k=0}^{\infty} k f(k; n, p) \\
 &= \sum_{k=1}^n \frac{k n!}{k!(n-k)!} p^k q^{n-k} \\
 &= np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{(n-1)-(k-1)} \\
 &= np \sum_{k'=0}^{n'} f(k'; n', p), \text{ where } k' = k - 1, n' = n - 1 \\
 &= np
 \end{aligned}$$

Gaussian distribution

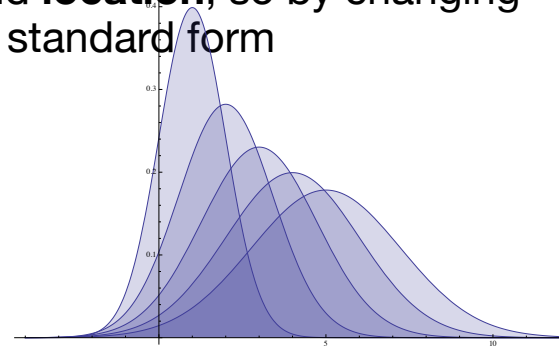
- The **normal** distribution – also known as **Gaussian** – is defined by the two-parameter probability density function

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$



- Actually, the two parameters give its **scale** and **location**, so by changing to a rescaled variable, all Gaussians have the standard form

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \text{ where } z = \frac{x - \mu}{\sigma}$$



Gaussian distribution

- Check the normalization, mean, and variance with handy formulas

$$I(a) = \int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

$$-\frac{dI(a)}{da} = \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a^3}}$$

- Probability in a range computed from CDF,

$$F(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-z'^2/2} dz' = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$F(b) - F(a) = \text{Prob}(a < Z < b)$$

Lognormal distribution

- If we change variables and substitute, obtain the **lognormal** density function

$$X = \log Y, \quad x = \log y,$$

$$\text{then } g(y) = \frac{p(x)}{|dy/dx|} = \frac{1}{y\sqrt{2\pi\sigma^2}} e^{-(\log y - \mu)^2 / (2\sigma^2)}$$

- The moments are more easily computed in original variables,

$$E[R] = E[e^r - 1] = e^{\mu + \sigma^2/2} - 1 \approx \mu + \sigma^2/2 + \dots,$$

$$\text{Var}(R) = E[(R - \bar{R})^2] = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) \approx \sigma^2(1 + 2\mu) + \dots$$

Poisson distribution

- Depends on two parameters, one integer and one real:

$$p(k; \lambda) = \frac{e^{-\lambda} \lambda^k}{k!}$$

- Application: probability to find exactly k "arrivals" or "events" during interval of time t .

$$p(k; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

Poisson distribution

- Mean

$$\mu = E[X] = e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} = \lambda e^{-\lambda} \sum_{k'=0}^{\infty} \frac{\lambda^{k'}}{k'!} = \lambda$$

- Second moment

$$E[X^2] = e^{-\lambda} \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} = \lambda + \lambda^2,$$

$$\sigma^2 = E[X^2] - \mu^2 = \lambda$$

Poisson distribution

- Limiting case of binomial distribution $n \rightarrow \infty, \quad np = \lambda$ fixed

$$p(k; \lambda) = \frac{e^{-\lambda} \lambda^k}{k!}$$

- Example: how many students in a room have a birthday today?

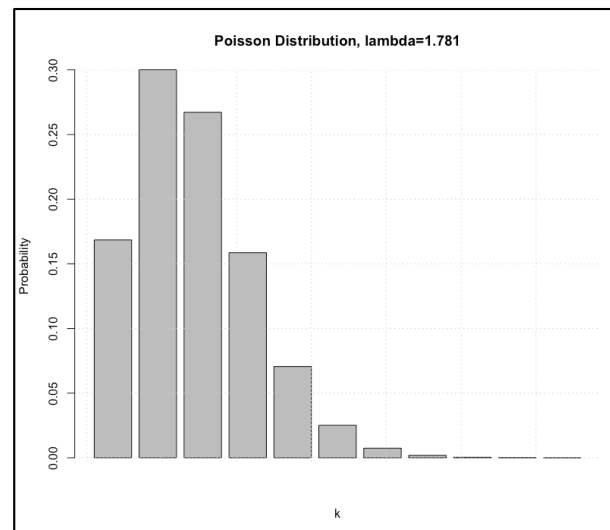
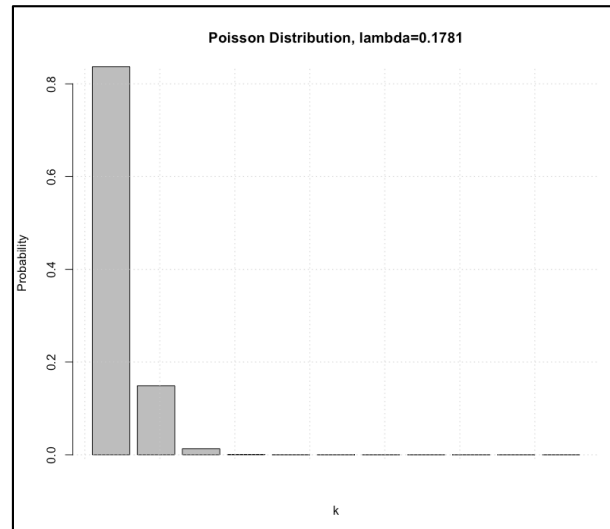
$$p = \frac{1}{365}, \quad n = 65$$

$$\lambda = np \approx 0.1781$$

Number k	Binomial probability	Poisson probability
0	0.8367	0.8369
1	0.1494	0.1490
2	0.0131	0.0133

Poisson distribution

- For small lambda, values are always decreasing in k .
- For $\lambda > 1$, there is a single maximum for k near λ
- Distribution is **skewed** with a long right-hand tail.



Poisson distribution

- A few more handy formulas for evaluating moments:

$$\frac{k}{k!} = \frac{k}{k(k-1)!} = \frac{1}{(k-1)!},$$

$$\frac{k^2}{k!} = \frac{k}{(k-1)!} = \frac{(k-1) + 1}{(k-1)!} = \frac{1}{(k-2)!} + \frac{1}{(k-1)!},$$

$$\frac{k^3}{k!} = \frac{k^2}{(k-1)!} = \frac{(k^2 - 3k + 2) + (3k - 3) + 1}{(k-1)!} = \frac{1}{(k-3)!} + \frac{3}{(k-2)!} + \frac{1}{(k-1)!}$$

- To simplify these ratios, add and subtract terms in the numerators that help simplify the denominators, always starting from the highest power of k and working down
- Is there a more general way to generate **all the moments** of a distribution?

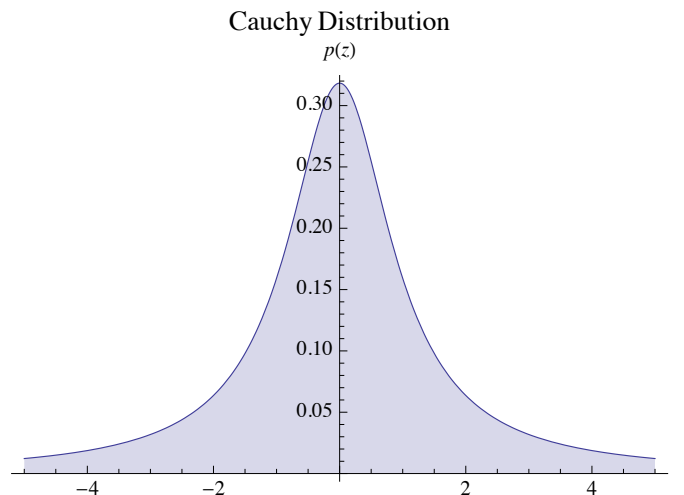
Fat tails

- The term "fat tails" is sometimes used to refer to distributions with a large kurtosis.
- It may also refer to distributions for which **higher moments diverge**, in which case the CLT may not hold.

- Example: power-law tails

- Cauchy distribution

$$p(x) = \frac{A}{\pi^2 A^2 + x^2} \sim \frac{A}{x^2}$$



- Variance and higher moments are infinite

Common distributions

Summary

- Common probability distributions used in finance include
 - Uniform
 - Binomial
 - Poisson
 - Normal (Gaussian)
 - Lognormal
- These have well-defined moments and interesting limiting cases
- Applications include option pricing, credit defaults, jump processes, and most models of asset pricing and forecasting.
- Choosing the right distribution depends on theory (sometimes) and on empirical observations (always).

Sums of random variables

Multiple random variables

- When there are several random variables, it is often useful to ask about the **distribution of their sum**.

$$S = X_1 + X_2 + \cdots + X_n$$

- In simple cases, the full probability distribution can be derived.
- In more complex cases, we only try to compute some of the **moments** of the full distribution, such as the mean and the variance.
- Easy to do using **linearity** of the expectation operator

$$E[S] = E[X_1] + E[X_2] + \cdots + E[X_n]$$

- Due to the CLT, this is often enough! If there are enough variables, and if the individual distributions in the sum are well-behaved, then the sum will be an (approximately) Gaussian variable...in which case the mean and variance describe it completely.

Application: portfolio returns

- Financial modeling applications include sums over assets and sums across time. Their **risk** depends on the **covariance**.

▪ **Example:** Let a portfolio consist of N assets and let

- w_i = portfolio weight in asset i ,
- R_i be the return on asset i ,
- $\mu_i = E[R_i]$, $\sigma_i^2 = \text{Var}(R_i)$

$$R_p = \sum_{i=1}^N w_i R_i = w_1 R_1 + w_2 R_2 + \cdots + w_N R_N$$

$$\mu_p = E[R_p] = \sum_{i=1}^N w_i E[R_i] = \sum_{i=1}^N w_i \mu_i$$

Correlated random variables

- No assumptions made about the individual return distributions except for the existence of the mean and variance.
- The portfolio risk, measured by its standard deviation, does involve the **covariance of pairs** of asset returns

$$\begin{aligned}
 \sigma_p^2 &= \text{Var}(R_p) = \text{E} \left[(R_p - \mu_p)^2 \right] = \text{E} \left[\left(\sum_{i=1}^N w_i (R_i - \mu_i) \right)^2 \right] \\
 &= \sum_{i=1}^N w_i^2 \text{E} \left[(R_i - \mu_i)^2 \right] + 2 \sum_{i < j}^N w_i w_j \text{E} \left[(R_i - \mu_i)(R_j - \mu_j) \right] \\
 &= \sum_{i=1}^N w_i^2 \text{Var}(R_i) + 2 \sum_{i < j}^N w_i w_j \text{Cov}(R_i, R_j) = \sum_{i=1}^N w_i^2 \sigma_i^2 + 2 \sum_{i < j}^N w_i w_j \sigma_i \sigma_j \rho_{ij}
 \end{aligned}$$

Correlated random variables: special cases

- Correlation = 0, variance equals sum of squares

$$\rho_{ij} = 0, \quad \sigma_p^2 = \sum (w_i \sigma_i)^2$$

- Correlation = 1, variance equals square of sum

$$\rho_{ij} = 1, \quad \sigma_p^2 = \left(\sum w_i \sigma_i \right)^2$$

- Correlation = 0, all weights and variances identical

$$\rho_{ij} = 0, w_i = \frac{1}{N}, \sigma_i = \sigma_0 \quad \implies \quad \sigma_p^2 = \frac{\sigma_0^2}{N}$$

Two random variables

- Two, however, is not a large number.

Let $S = X_1 + X_2$ be the sum of two independent random variables with density function $p_1(x_1)$ and $p_2(x_2)$. Then

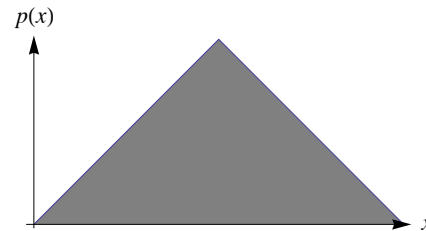
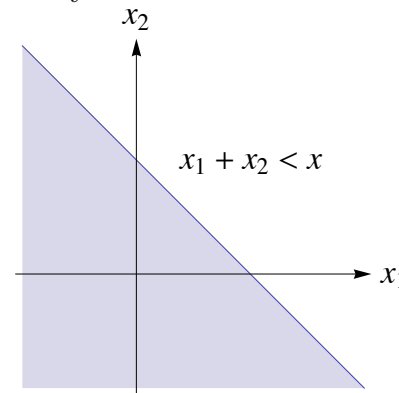
$$\begin{aligned} F(x) &= \text{Prob}(X_1 + X_2 < x) = \int \int_A p_1(x_1)p_2(x_2)dx_1dx_2 \\ &= \int_{x_1=-\infty}^{x_1=\infty} \int_{x_2=-\infty}^{x_2=x-x_1} p_1(x_1)p_2(x_2)dx_1dx_2 \end{aligned}$$

So

$$p(x) = \frac{dF(x)}{dx} = \int_{-\infty}^{\infty} p_1(x_1)p_2(x - x_1)dx_1,$$

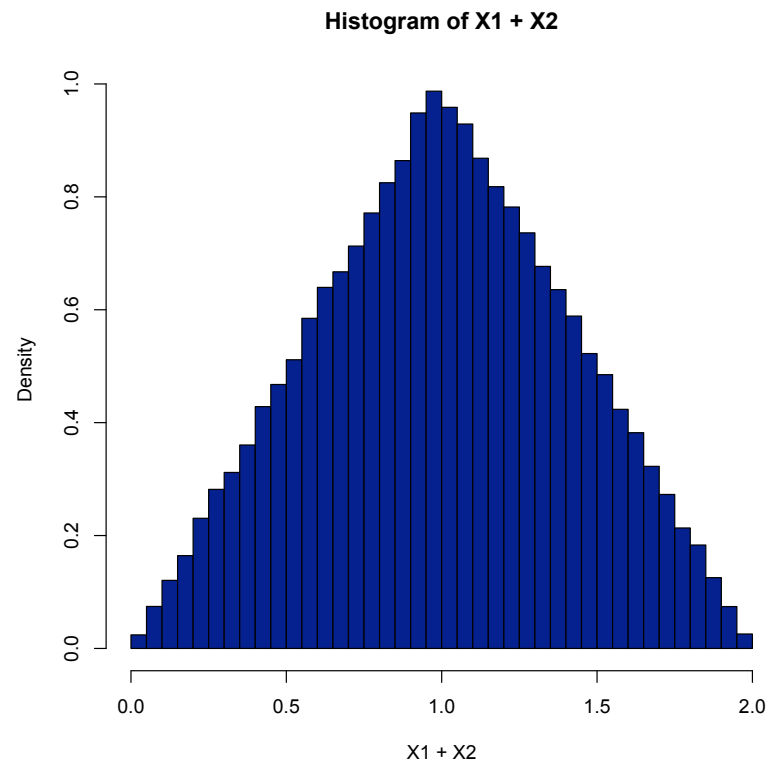
which defines the *convolution* of p_1 and p_2 . In the case of the uniform distribution

$$p(x) = \begin{cases} x & x \in [0, 1] \\ 2 - x & x \in [1, 2] \\ 0 & \text{otherwise} \end{cases}$$



Two random variables: simulation

```
X1 <- runif(1e5)  # X1 and X2 drawn from uniform distribution on [0,1]  
X2 <- runif(1e5)  
hist(X1+X2, probability=TRUE, breaks=50)
```



Binomial distribution (the easy way)

- We can view the number of "successes" as a **sum of random variables**, one for each of the Bernoulli trials,

$$\text{Let } S = X_1 + X_2 + \dots + X_n.$$

$$\begin{aligned} \text{Then } E[S] &= E[X_1] + E[X_2] + \dots + E[X_n] \\ &= np. \end{aligned}$$

- This approach builds up results, using the expectations of one or two of the random variables at a time and the fact that the X 's are **independent** and **identically distributed**.

$$E[X_1] = p \cdot 1 + q \cdot 0 = p,$$

$$E[X_1^2] = p \cdot 1^2 + q \cdot 0^2 = p,$$

$$E[X_1 X_2] = p^2 \cdot 1 \cdot 1 + pq \cdot 1 \cdot 0 + qp \cdot 0 \cdot 1 + q^2 \cdot 0 \cdot 0 = p^2$$

Binomial distribution

- This approach makes the variance especially easy to compute:

$$\begin{aligned}
 \sigma^2 &= E[(S - \mu)^2] \\
 &= E[(X_1 + X_2 + \dots + X_n)^2] - \mu^2 \\
 &= n E[X_1^2] + n(n-1) E[X_1 X_2] - \mu^2 \\
 &= np + n(n-1)p^2 - (np)^2, \text{ since } E[X_i X_j] = \begin{cases} p, & \text{if } i = j \\ p^2, & \text{if } i \neq j \end{cases} \\
 &= np(1-p) = npq,
 \end{aligned}$$

$$\sigma = \sqrt{npq}.$$

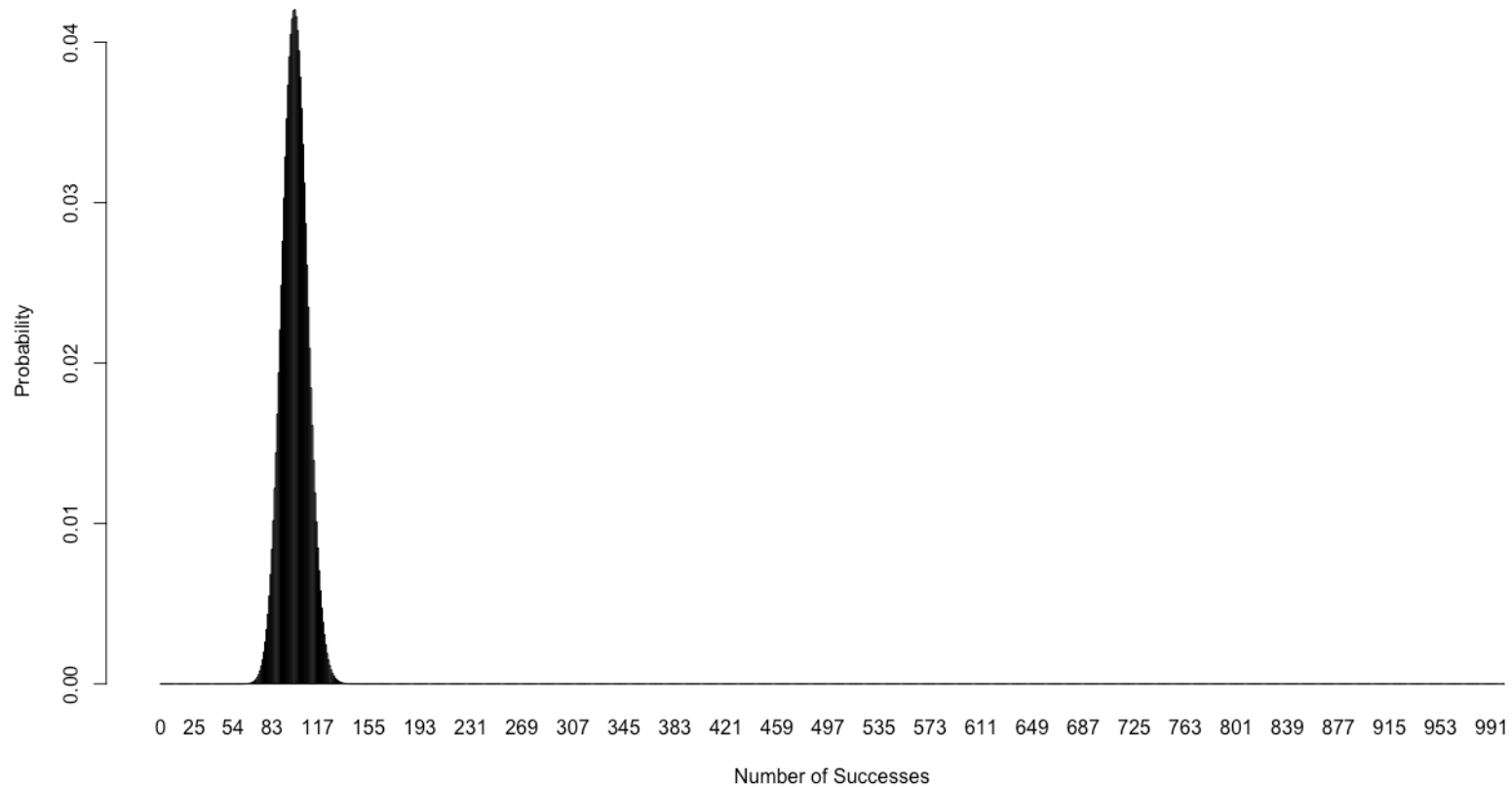
Binomial distribution

- In numerical calculations, factorial factors quickly overflow machine limits
- R functions:
 - choose(n,k)
 - dbinom(k,n,p)

```
nlist <- c(1,2,5,10,20,50,100,1000)
p <- 0.1
for (n in nlist) {
  k <- 0:n
  f <- dbinom(k,n,p)
  barplot(f,names=k,
    xlab="Number of Successes",ylab="Probability",
    main=paste("Binomial Distribution, p=",p," n=",n))
  readline()
}
```

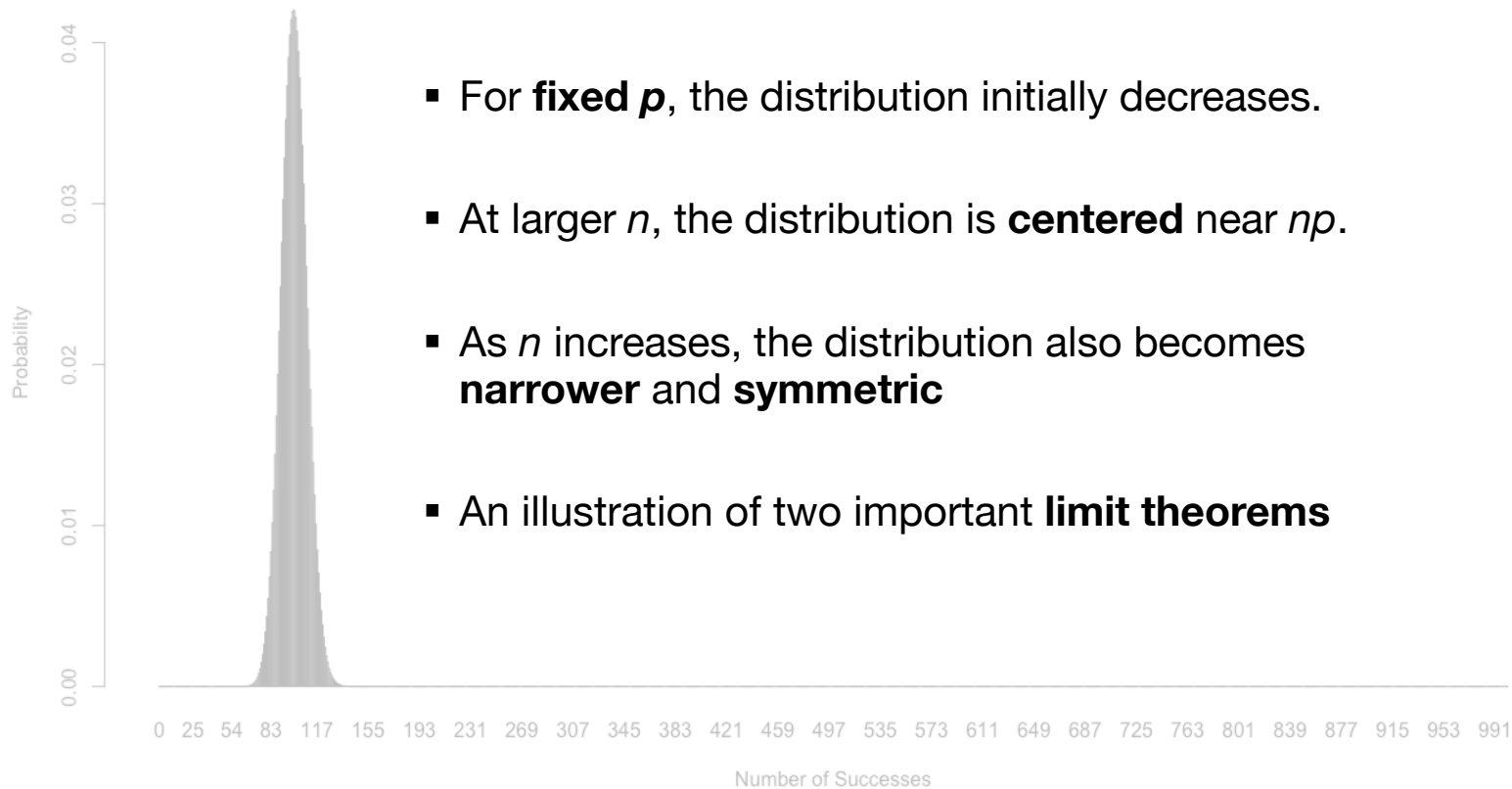

Binomial distribution, $p=10\%$, $n=1,2,\dots,1000$

Binomial Distribution, $p=0.1$ $n=1000$



Binomial distribution

Binomial Distribution, $p = 0.1$ $n = 1000$



- For **fixed p** , the distribution initially decreases.
- At larger n , the distribution is **centered** near np .
- As n increases, the distribution also becomes **narrower** and **symmetric**
- An illustration of two important **limit theorems**

Binomial distribution

An illustration of two important **limit theorems**

- **Law of Large Numbers (LLN):**

- As n increases, the probability that the mean deviates from np goes to zero.

- **Central Limit Theorem (CLT):**

- As n increases for fixed p , the distribution approaches Gaussian (or Normal)

Binomial distribution

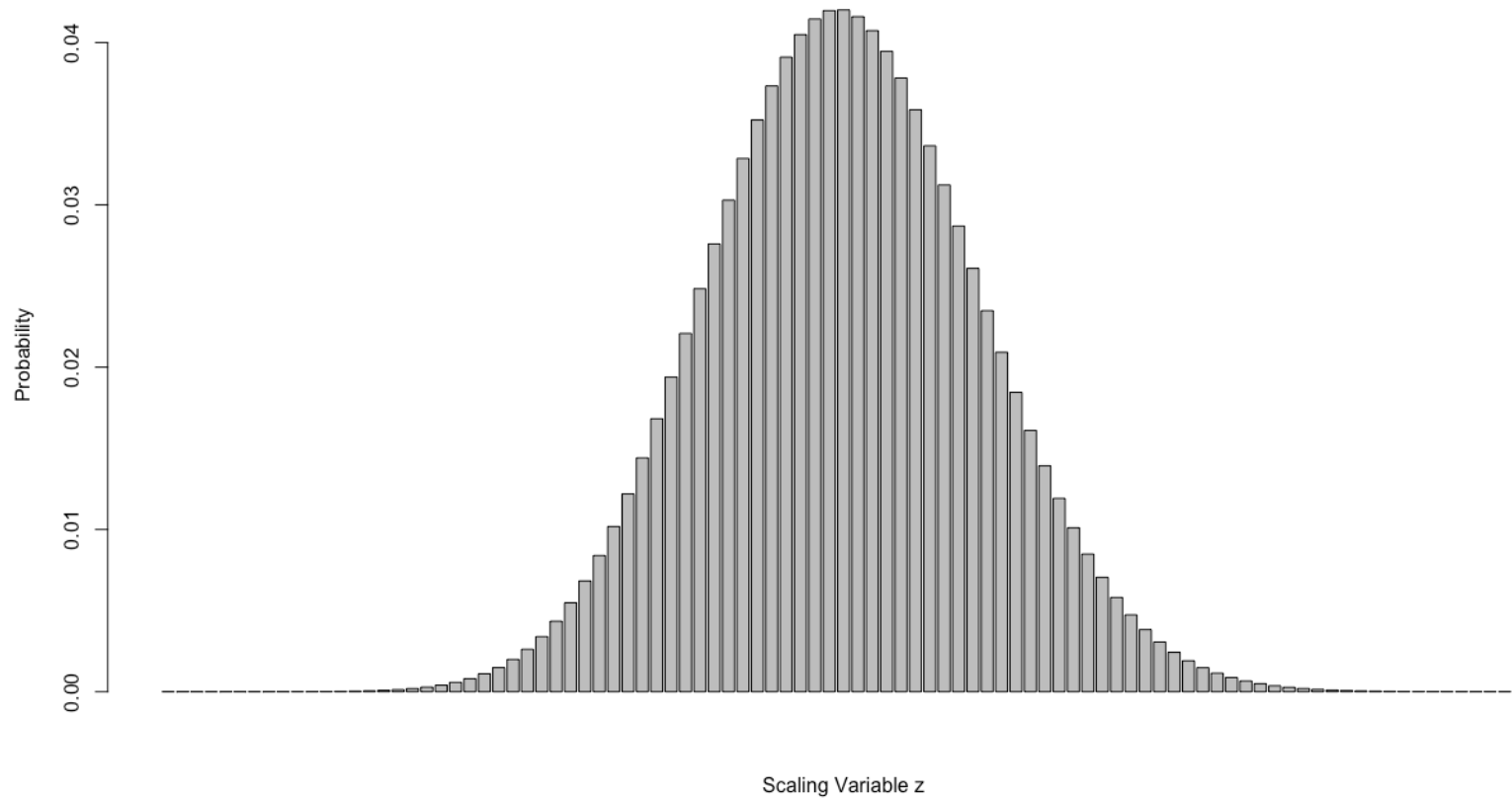
- Define the **scaling variable** $z = z_k = \frac{k - np}{\sqrt{npq}}$

```
nlist <- c(1,2,5,10,20,50,100,1000)
p      <- 0.1
zmax   <- 5

for (n in nlist) {
  k   <- 0:n
  z   <- (k - n*p)/sqrt(n*p*(1-p))
  zi  <- (abs(z)<=zmax)
  f   <- dbinom(k,n,p)
  barplot(f[zi], xlab="Scaling Variable z", ylab="Probability",
    main=paste("Binomial Distribution, p=",p," n=",n))
  readline()
}
```

Scaled binomial distribution

Binomial Distribution, $p = 0.1$ $n = 1000$



Binomial distribution

- In terms of the scaling variable, the limiting distribution is **Gaussian**

$$z = z_k = \frac{k - np}{\sqrt{npq}}$$

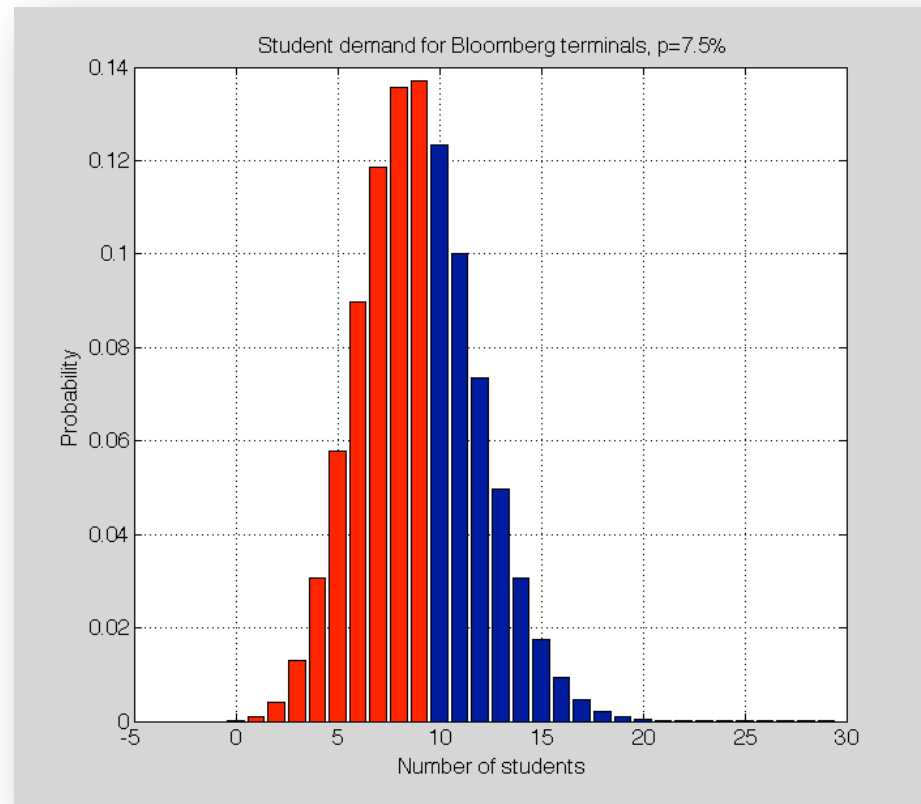
$$f(k; n, p) \approx f(z_k) = \frac{1}{\sqrt{2\pi}} e^{-z_k^2/2}$$

Example: Bloomberg terminals

- Parameters:
 - 120 students
 - 9 terminals
 - Demand independently with probability 7.5%
- Cumulative probability 58.75% that 9 or fewer students will want to use at same time.

```
n <- 120
k <- 0:9
p <- 0.075

sum(dbinom(k,n,p))
[1] 0.587475
```



Sums of random variables

Summary

- A set of individual random variables can be added to create a new random variable, their sum.

$$S_n = X_1 + X_2 + \cdots + X_n$$

- The full probability distribution can be computed using convolutions (which is hard), but we will usually need only the moments (which is easy).
- The **moments** of this new random variable can be easily computed using **linearity**.
- If the variables are **independent** and **identically distributed (IID)**, then the sum is **normally distributed** as $n \rightarrow \infty$.
- When the variables are ordered in **time**, the sequence of sums forms a **stochastic process**.

From moment generating functions to the CLT

Characteristic function

- The **characteristic function** is defined so that its derivatives **generate** all the moments of the distribution

$$\tilde{f}(t) \equiv E[e^{itX}], \quad E[X^\ell] = (-i)^\ell \frac{d^\ell}{dt^\ell} \tilde{f}(t) \Big|_{t=0}$$

- It is useful whenever the function and its derivatives can be computed in closed form. For the binomial distribution,

$$\begin{aligned} \tilde{f}(t) &\equiv E[e^{itX}] = \sum_{k=0}^{\infty} e^{itk} \text{Prob}(X = k) \\ &= \sum_{k=0}^n e^{itk} \binom{n}{k} p^k q^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} (pe^{it})^k q^{n-k} \\ &= (pe^{it} + q)^n \end{aligned}$$

Hence

$$\begin{aligned} E[1] &= \tilde{f}(0) = (p + q)^n = 1, \\ E[X] &= -i\tilde{f}'(0) = np, \\ E[X^2] &= -\tilde{f}''(0) = npq + n^2p^2, \text{ etc.} \end{aligned}$$

Characteristic function

- Continuous distributions: Fourier transform of density

$$\tilde{p}(t) \equiv \int_{-\infty}^{\infty} e^{itx} p(x) dx = \sum_{\ell=0}^{\infty} \frac{(it)^{\ell}}{\ell!} E[X^{\ell}], \quad \mu_{\ell} = E[X^{\ell}] = (-i)^{\ell} \left. \frac{d^{\ell}}{dt^{\ell}} \tilde{p}(t) \right|_{t=0}$$

Especially useful for **sums of random variables**: $Y = X_1 + X_2$,

- Density function of the sum
= **convolution** of the individual functions $p(y) = \int p_1(x_1)p_2(y - x_1)dx_1$,
- Fourier transform of a convolution
= **product** of the Fourier transforms $\tilde{p}(t) = \tilde{p}_1(t)\tilde{p}_2(t)$

Normal distribution

- Gaussians are very special
 - The Fourier transform of a Gaussian is also a Gaussian

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}, \quad \tilde{p}(t) = e^{-\sigma^2 t^2/2 + i\mu t}$$

Normal distribution

- The sum of N independent Gaussian random variables is itself a Gaussian RV with its own mean and variance:

$$\begin{aligned}
 \tilde{p}(t) &= \tilde{p}_1(t) \tilde{p}_2(t) \cdots \tilde{p}_N(t) \\
 &= \exp \left[-\frac{\sigma_1^2 t^2}{2} + i\mu_1 t \right] \exp \left[-\frac{\sigma_2^2 t^2}{2} + i\mu_2 t \right] \cdots \\
 &= \exp \left[-\frac{(\sigma_1^2 + \cdots + \sigma_N^2) t^2}{2} + i(\mu_1 + \cdots + \mu_N) t \right] \\
 &= \exp \left[-\frac{\hat{\sigma}^2 t^2}{2} + i\hat{\mu} t \right],
 \end{aligned}$$

where

$$\hat{\sigma}^2 = \sum \sigma_i^2, \quad \hat{\mu} = \sum \mu_i$$

Scaling for Gaussians

- The sum of N independent, **identical** Gaussian random variables is a Gaussian RV with

$$\mu \rightarrow \hat{\mu} = N\mu,$$

$$\sigma \rightarrow \hat{\sigma} = \sqrt{N}\sigma$$

Cumulant expansion

- Since Fourier transforms multiply, their logarithms **add**. Define the **cumulant** expansion,

$$\tilde{p}(t) \equiv \exp \left[\sum_{\ell=0}^{\infty} \frac{(it)^{\ell}}{\ell!} C_{\ell} \right], \quad C_n = (-i)^n \frac{d^n}{dt^n} \log \tilde{p}(t) \Big|_{t=0}$$

- The first few terms are

$$C_1 = \langle X \rangle,$$

$$C_2 = \langle X^2 \rangle - \langle X \rangle^2,$$

$$C_3 = \langle X^3 \rangle - 3\langle X \rangle \langle X^2 \rangle + 2\langle X \rangle^3,$$

$$C_4 = \langle X^4 \rangle - 3\langle X^2 \rangle^2 - 4\langle X \rangle \langle X^3 \rangle + 12\langle X \rangle^2 \langle X^2 \rangle - 6\langle X \rangle^4$$

- For a Gaussian, all cumulants are exactly zero for $n > 2$.

Central limit theorem

- Consider the cumulants for a **sum of N independent, identically-distributed** random variables. Since cumulants add,

$$\hat{C}_n = NC_n,$$

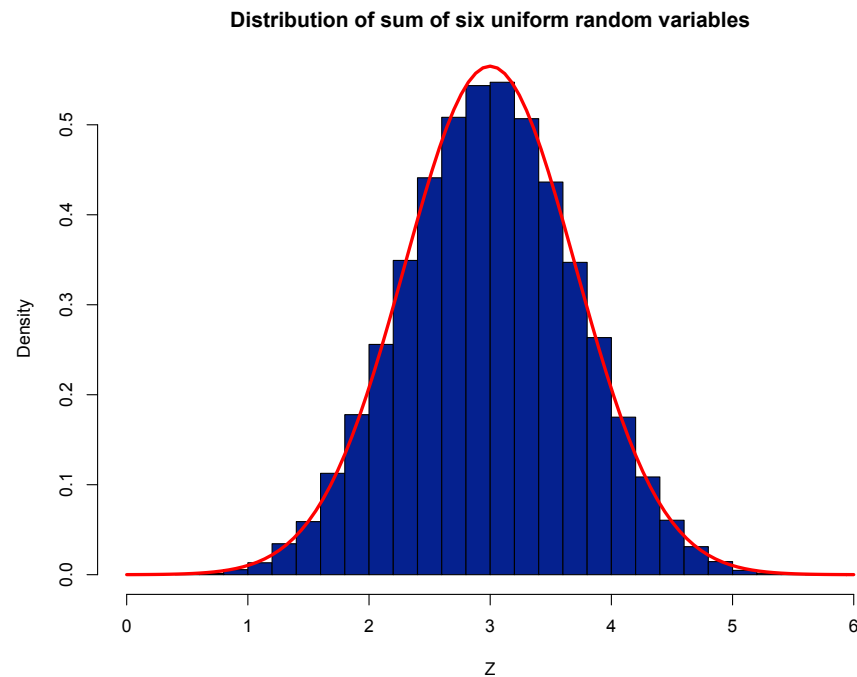
- Then the dimensionless, normalized cumulants have a power-law dependence on N :

$$\frac{\hat{C}_n}{\hat{\sigma}^n} = \frac{1}{N^{n/2-1}} \left(\frac{C_n}{\sigma^n} \right)$$

- As $N \rightarrow \infty$, all cumulants vanish for $n > 2$, and the distribution for the sum **approaches a Gaussian**.
- However, this provides no information about the rate of convergence for different values of x ; i.e., tails vs. center.

Six random variables: simulation

```
#Add 6 random variables drawn from uniform distribution on [0,1]
Z <- rowSums(matrix(runif(6e5),ncol=6))
hist(Z,probability=TRUE, breaks=30, col="#05218F",xlim=c(0,6),
     main="Distribution of sum of six uniform random variables")
curve(dnorm(x,mean=mean(Z),sd=sd(Z)),0,6,col="red",lwd=3,add=T)
```



Characteristic functions and the CLT

Summary

- The **characteristic function** of a probability distribution gives a compact formula for generating all of the moments.
- For continuous random variables, it is given by Fourier transforms, which have simple properties when applied to sums.
- Gaussians are special
 - The Fourier transform of a Gaussian is also a Gaussian.
 - A sum of Gaussian random variables is also a Gaussian random variable.
 - The central limit theorem (CLT) shows that Gaussians are **universal**: adding together a large number of **any** IID random variables approaches a Gaussian distribution*.

**Conditions apply. You mileage may vary.*